

Variance Reduction Methods (Part II)

SIE 449/549: Optimization for Machine Learning

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Strongly Convex Function

- ▶ f is **convex** if and only if

$$f(y) \geq f(x) + \nabla f(x)^T (y - x) \quad \text{for all } x, y$$

- ▶ f is **strictly convex** if and only if

$$f(y) > f(x) + \nabla f(x)^T (y - x) \quad \text{for all } x, y$$

- ▶ f is **strongly convex** if and only if

$$f(y) \geq f(x) + \nabla f(x)^T (y - x) + \frac{\mu}{2} \|x - y\|^2 \quad \text{for all } x, y$$

we say f is μ -strongly convex

Strongly Convex Function

- If f is twice continuously differentiable, then

$$f \text{ is } \mathbf{convex} \iff \nabla^2 f(x) \succeq 0 \text{ for all } x$$

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Strongly Convex Function

- ▶ If the objective function is **convex**, then any local minimum is a global minimum
- ▶ If the objective function is **strictly convex**, then the minimum is unique
- ▶ If the objective function is **strongly convex**, then the minimum is unique

Method	Convex function	Strongly convex function
Gradient Descent (GD)	$\mathcal{O}(1/\epsilon)$	$\mathcal{O}(\log(1/\epsilon))$
Accelerated GD	$\mathcal{O}(1/\sqrt{\epsilon})$	$\mathcal{O}(\log(1/\epsilon))$
Proximal GD	$\mathcal{O}(1/\epsilon)$	$\mathcal{O}(\log(1/\epsilon))$
Subgradient	$\mathcal{O}(1/\epsilon^2)$	$\mathcal{O}(1/\epsilon)$
SG	$\mathcal{O}(1/\epsilon^2)$	$\mathcal{O}(1/\epsilon)$
SAG/SAGA	$\mathcal{O}(1/\epsilon)$	$\mathcal{O}(\log(1/\epsilon))$

- ▶ Suppose an algorithm has a **linear convergence rate** $\mathcal{O}(\beta^k)$, where $\beta \in (0, 1)$, then the rate in terms of ϵ is

Stochastic Variance Reduced Gradient (SVRG)

- ▶ Consider the following convex optimization problem:

$$\min \frac{1}{n} \sum_{i=1}^n f_i(x)$$

- ▶ SAG and SAGA were introduced to improve convergence rate of SG
- ▶ However, both methods require storage of all gradients

Goal: Employs variance reduction without the need to store the intermediate gradients

SVRG*

Algorithm 1 SVRG

Parameters: update frequency m and stepsize α

Initialization: pick $x_0 \in \mathbb{R}^n$ arbitrarily and set $\tilde{x}_0 = x_0$

for $s = 1, 2, \dots$ **do**

$$\tilde{x} = \tilde{x}_{s-1}$$

$$\tilde{g} = \frac{1}{n} \sum_{i=1}^n \nabla f_i(\tilde{x})$$

$$x_0 = \tilde{x}$$

for $t = 1, 2, \dots, m$ **do**

Randomly pick $i \in \{1, \dots, n\}$ and update

$$x_t = x_{t-1} - \alpha (\nabla f_i(x_{t-1}) - \nabla f_i(\tilde{x}) + \tilde{g})$$

end for

Option I: $\tilde{x}_s = x_m$

Option II: $\tilde{x}_s = x_t$ for randomly chosen $t \in \{1, \dots, m\}$

end for

*Johnson, Rie, and Tong Zhang. "Accelerating stochastic gradient descent using predictive variance reduction." Advances in neural information processing systems 26 (2013).

Unbiasedness and Variance Reduction

- ▶ We have $\tilde{g} = \frac{1}{n} \sum_{i=1}^n \nabla f_i(\tilde{x})$
- ▶ Gradient estimator: $\nabla f_i(x_{t-1}) - \nabla f_i(\tilde{x}) + \tilde{g}$

- ▶ When both $\tilde{x}$ and x_t converge to the same parameter x^* , then $\tilde{g} \rightarrow 0$

- ▶ Therefore if $\nabla f_i(\tilde{x}) \rightarrow \nabla f_i(x^*)$, then:

Lipschitz and Expectation Properties

Lemma 1

If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a differentiable convex function with L -Lipschitz continuous gradient, then the following conditions are equivalent for all x, y :

- (i) $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$
- (ii) $f(y) \leq f(x) + \nabla f(x)^T(y - x) + \frac{L}{2}\|x - y\|^2$
- (iii) $\langle \nabla f(x) - \nabla f(y), x - y \rangle \leq L\|x - y\|^2$
- (iv) $\|\nabla f(x) - \nabla f(y)\|^2 \leq L(\nabla f(x) - \nabla f(y))^T(x - y)$
- (v) $\|\nabla f(x) - \nabla f(y)\|^2 \leq 2L(f(x) - f(y) - \nabla f(x)^T(x - y))$

Lemma 2

$$\mathbb{E}[\|\xi - \mathbb{E}[\xi]\|^2] = \mathbb{E}[\|\xi\|^2] - \|\mathbb{E}[\xi]\|^2 \leq \mathbb{E}[\|\xi\|^2].$$

SVRG: Variance Reduction

- Consider SVRG algorithm. Define $g = \nabla f_i(x_t) - \nabla f_i(\tilde{x}) + \tilde{g}$, where $\tilde{g} = \frac{1}{n} \sum_{i=1}^n \nabla f_i(\tilde{x})$, then we have

$$\mathbb{E}[\|g - \nabla f(x_t)\|^2] \leq 4L(f(x_t) - f(x^*) + f(\tilde{x}) - f(x^*))$$

Convergence Analysis

Theorem 3

Consider SVRG algorithm. Assume f is μ -strongly convex and each f_i has a Lipschitz gradient. Assume m is sufficiently large so that

$$\beta = \frac{1}{\mu\alpha(1-2L\alpha)m} + \frac{2L\alpha}{1-2L\alpha} < 1$$

then,

$$\mathbb{E}[f(\tilde{x}_s)] - f(x^*) \leq \beta^s (f(\tilde{x}_0) - f(x^*))$$

- ▶ **Number of inner iterations:**
- ▶ **Complexity:**
- ▶ Unlike SG, the α for SVRG does not have to decay
- ▶ SVRG explicitly reduces the variance of SG

SVRG++[†]

- Consider $\min F(x) \triangleq \frac{1}{n} \sum_{i=1}^n f_i(x) + h(x)$, where f_i s are convex and differentiable, h is convex but possibly nonsmooth

Algorithm 2 SVRG++

Initialization: pick $\tilde{x}_0 \in \mathbb{R}^n$ arbitrarily
for $s = 1, 2, \dots, S$ **do**
 $\tilde{x} = \tilde{x}_{s-1}$
 $\tilde{g} = \frac{1}{n} \sum_{i=1}^n \nabla f_i(\tilde{x})$
 $m_s = 2^s m_0$
 for $t = 1, 2, \dots, m_s$ **do**
 Randomly pick $i \in \{1, \dots, n\}$ and update
 $g = \nabla f_i(x_{t-1}) - \nabla f_i(\tilde{x}) + \tilde{g}$
 $x_t = \text{prox}_{h, \alpha}(x_{t-1} - \alpha g)$
 end for
 $\tilde{x}_s = \frac{1}{m_s} \sum_{t=1}^{m_s} x_t$
 $x_0 = x_{m_s}$
end for

[†]Allen-Zhu, Zeyuan, and Yang Yuan. "Improved SVRG for non-strongly-convex or sum-of-non-convex objectives." International conference on machine learning. PMLR, 2016.

Convergence Analysis

Theorem 4

Consider SVRG++ and assume each f_i is convex with a Lipschitz gradient. Choose $\alpha = 1/(7L)$, then,

$$\mathbb{E}[F(\tilde{x}_s)] - F(x^*) \leq \mathcal{O} \left(\frac{F(x_0) - F(x^*)}{2^s} + \frac{L\|x_0 - x^*\|^2}{2^s m_0} \right)$$

► **Complexity:**

► Compare to SAG/SAGA: